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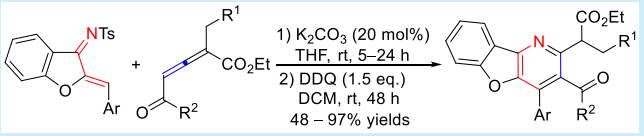
Base-Catalyzed Sequential 1,4-Addition/Intramolecular Cyclization/Aromatization Reaction: Synthesis of Benzofuro[3,2-*b*]pyridines

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ABSTRACT: An efficient K_2CO_3-catalyzed 1,4-addition/intramolecular cyclization/aromatization reaction of aurone-derived 1-azadienes with trisubstituted allenoates has been developed, giving a series of benzofuro[3,2-*b*]pyridines in moderate to excellent yields. The reaction proceeded efficiently under an air atmosphere without the use of transition metal catalysts. This protocol provides a concise approach to benzofuro[3,2-*b*]pyridines.



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Polyheterocyclic frameworks are ubiquitous in natural products and pharmacologically active molecules.[1] Among them, benzofuro[3,2-*b*]pyridines (BFPs) (Figure 1)

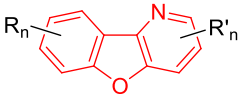
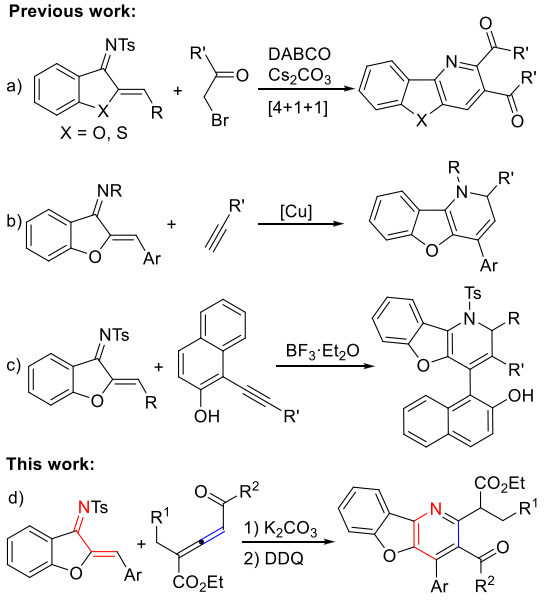


Figure 1. Benzofuro[3,2-*b*]pyridine motif.

represent an important class of heterocyclic compounds.[2−5] Various strategies for their construction have been established.[6−9] However, some of these methods suffer from multistep procedures and the limitation of a narrow substrate range. Therefore, it still is highly desirable to develop a simple method for the construction of this important scaffold.

In recent years, aurone-derived 1-azadienes have been used as highly reactive intermediates in organic synthesis, due to the driving force of aromatization.[7−10] With these aurone-derived 1-azadienes as the starting material, several elegant methods for the construction of benzofuro[3,2-*b*]pyridine were established. In 2019, Chen, Jiang and co-workers developed an intriguing base-promoted [4+1+1] annulation reaction of aurone-derived 1-azadienes with α-bromo carbonyls, providing a complex benzofuro[3,2-*b*]pyridines scaffold in an efficient manner (Scheme 1a).[7] By using a copper catalyst, Zhou achieved formal [4+2] cycloaddition of aurone-derived 1-azadienes with terminal alkynes, which elegantly produced 1,2-dihydrobenzofuran[3,2-*b*]pyridines through a complex nucleophilic addition/cyclization/isomerization process (Scheme 1b).[8] In 2021, Jiang, Tu, Hao, and co-workers presented a Lewis acid-catalyzed [2+2] cycloaddition/retroelectrocyclization/1,6-addition relay of aurone-derived 1-azadienes and 1-alkynylnaphthalen-2-ols, resulting in the regio- and diastereoselective synthesis of 1,2-dihydrobenzofuro[3,2-*b*]pyridine with a chiral carbon center and an axial chirality in good yields

Scheme 1. Construction of Benzofuro[3,2-*b*]pyridine Scaffolds



(Scheme 1c).[9] According to previous studies,[11] it was known that the α-hydrogen of allenoates and allene ketones has a certain acidity, which allows allenoates and allene ketones to be deprotonated and thus act as nucleophiles. On the basis of this

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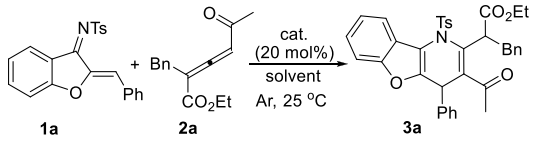
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feature, we investigated the base-assisted reaction of allenoates with aurone-derived 1-azadienes, and we herein describe an efficient K_2CO_3-catalyzed 1,4-addition/intramolecular cyclization/aromatization reaction of aurone-derived 1-azadienes with trisubstituted allenoates, giving a series of benzofuro[3,2-*b*]pyridines (Scheme 1d).

At the beginning of our study, we optimized the reaction of 1-azadiene **1a** and trisubstituted allenoate **2a** in CH_2Cl_2 under an argon atmosphere. As shown in Table 1, several LB catalysts

Table 1. Optimization of the Reaction Conditions[a]



1a 2a 3a

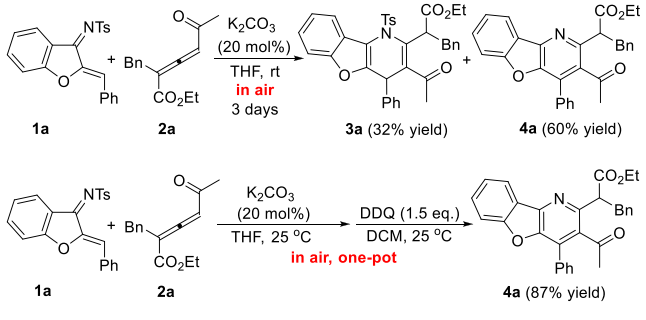
entry	catalyst	solvent	*t* (h)	yield (%)[b]
1	PPh_3	DCM	24	NR[c]
2	$MePPh_2$	DCM	24	messy
3	DABCO	DCM	24	NR[c]
4	DMAP	DCM	24	23
5	K_2CO_3	DCM	30	81
6	Na_2CO_3	DCM	24	NR[c]
7	Cs_2CO_3	DCM	4	52
8	K_2CO_3	THF	9	91
9	K_2CO_3	Tol	30	89
10	K_2CO_3	MeCN	9	49
11	K_2CO_3	MTBE	12	91

[a]All reactions of **1a** (0.10 mmol), **2a** (0.11 mmol), and a catalyst (20 mol %) were performed in 1 mL of solvent at 25 °C. [b]Isolated yield. [c]No reaction.

were screened (entries 1−4), but only 4-(dimethylamino)-pyridine (DMAP) led to the corresponding product **3a** in 23% yield (entry 4). Subsequently, we turned our attention to inorganic bases. To our delight, in the presence of 20 mol % K_2CO_3, product **3a** was obtained in 81% yield (entry 5). Na_2CO_3 did not catalyze this reaction (entry 6). With Cs_2CO_3 as the catalyst, product **3a** was obtained in 52% yield (entry 7). With the use of K_2CO_3 as the catalyst, several solvents were then screened (entries 8−11), indicating that tetrahydrofuran was the optimal solvent for this reaction (entry 8). The structure of product **3a** has been unambiguously confirmed by X-ray diffraction analysis.

We accidentally found that when the model reaction mixture was placed in air, product **3a** was obtained in 32% yield after 3 days, and the corresponding aromatization product **4a** was also isolated in 60% yield (Scheme 2). This triggered our

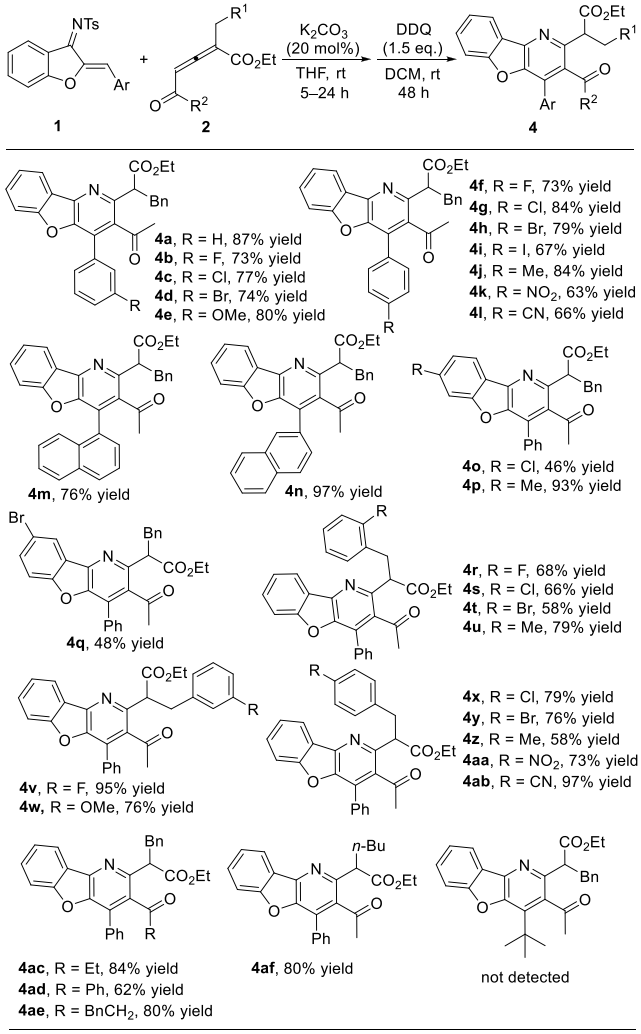
Scheme 2. Reaction in Air and One-Pot Synthesis of 4a



exploration of a one-pot strategy for the synthesis of aromatized product **4a** (Scheme 2). After a concise screening, we found that 1,2-dichloro-4,5-dicyanobenzoquinone (DDQ) was productive for this aromatization. After the 1,4-addition/intramolecular cyclization process had reached completion, 1.5 equiv of DDQ was added and product **4a** was produced in 87% yield in one pot.

To study the scope of this process, we investigated the reaction of various 1-azadienes **1** and different trisubstituted allenoates **2**, as shown in Table 2. The results indicated that

Table 2. Scope of Aurone-Derived 1-Azadienes and Trisubstituted Allenoates[a]



[a]The reactions of **1** (0.10 mmol), **2a** (0.11 mmol), and K_2CO_3 (20 mol %) were performed in THF (1.0 mL) at 25 °C for 5−24 h, and then DDQ (1.5 equiv) and DCM (1.0 mL) were added. The mixture was stirred at 25 °C for 48 h. The dr value of all products was >20:1, which was determined by ^{1}H NMR.

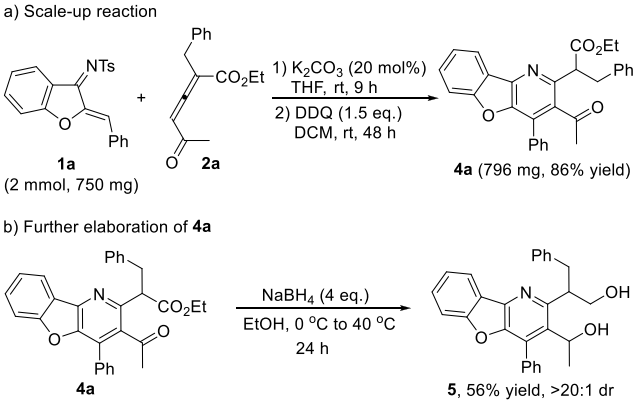
the position, steric properties, and electronic properties of the substituents on the aromatic ring of 1-azadienes **1** and the benzene ring of allenoates **2** had little effect on the reactivities. When the R^1 group was an alkyl group, the high reactivity was maintained, and product **4af** was formed in 80% yield. The impact of the R^2 group of the allenoates **2** linked to the ketone carbonyl on the reaction was also explored. To our delight, alkyl groups such as ethyl, phenethyl, and phenyl groups in

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allenoates **2** had an unremarkable impact on the reactivities. In addition, moderate to good results were gained with 1-azadienes **1** bearing electron-donating and -withdrawing groups at the benzofuran moiety (products **4o**−**4q**). Unfortunately, when the Ar group in 1-azadiene **1** was *tert*-butyl, the reaction did not work.

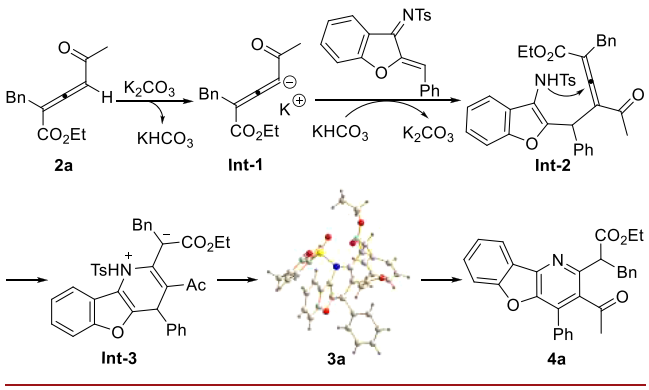
As indicated in Scheme 3, when this reaction was performed with 2 mmol (750 mg) of aurone-derived 1-azadiene **1a**, the reaction still worked well to produce product **4a** in 86% yield. There was an only slight loss of yield, compared with the reaction at a 0.1 mmol scale. Reduction of product **4a** with sodium borohydride gave glycol product **5** in 56% yield with >20:1 dr.

Scheme 3. Scale-up Reaction and Further Elaboration



The proposed reaction mechanism for the reaction is shown in Scheme 4. Initially, trisubstituted allenoate **2a** is deprotonated in the presence of K_2CO_3 to provide allene carbanion **Int-1**. Subsequent 1,4-addition of intermediate **Int-1** to 1-azadiene **1a** led to the formation of tetrasubstituted allenoate intermediate **Int-2**. Intramolecular cyclization of **Int-2** produced intermediate **Int-3**, which underwent a proton transfer to give 1,4-dihydrobenzofuro[3,2-*b*]pyridine **3a**. The succeeding aromatization occurred in the presence of DDQ to provide the final product, benzofuro[3,2-*b*]pyridine **4a**.

Scheme 4. A Plausible Reaction Mechanism



In summary, we have disclosed a concise one-pot synthesis of benzofuro[3,2-*b*]pyridines through an efficient K_2CO_3-catalyzed 1,4-addition/intramolecular cyclization/aromatization sequence. A series of benzofuro[3,2-*b*]pyridines were obtained in moderate to excellent yields. This approach features cheap K_2CO_3 as a catalyst, mild reaction conditions, and a wide range of substrates.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02357.

Experimental procedure, characterization data, NMR spectra, and X-ray crystallographic data (PDF)

Accession Codes

CCDC 2089438 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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REFERENCES

(1) For selected reviews, see: (a) Gomtsyan, A. Heterocycles in drugs and drug discovery. *Chem. Heterocycl. Compd.* **2012**, *48*, 7−10. (b) Taylor, A. P.; Robinson, R. P.; Fobian, Y. M.; Blakemore, D. C.; Jones, L. H.; Fadeyi, O. Modern advances in heterocyclic chemistry in drug discovery. *Org. Biomol. Chem.* **2016**, *14*, 6611−6637. (c) Chand, K.; Rajeshwari; Hiremathad, A.; Singh, M.; Santos, M. A.; Keri, R. S. A review on antioxidant potential of bioactive heterocycle benzofuran: Natural and synthetic derivatives. *Pharmacol. Rep.* **2017**, *69*, 281−295. (d) Keri, R. S.; Chand, K.; Budagumpi, S.; Balappa Somappa, S.; Patil, S. A.; Nagaraja, B. M. An overview of benzo[*b*]thiophene-based medicinal chemistry. *Eur. J. Med. Chem.* **2017**, *138*, 1002−1033.

(2) Liu, J.-Q.; Wang, C.-F.; Peng, X.-R.; Qiu, M.-H. New alkaloids from the fruiting bodies of Ganoderma sinense. *Nat. Prod. Bioprospect.* **2011**, *1*, 93−96.

(3) For selected examples, see: (a) Kwon, H. B.; Park, C.; Jeon, K. H.; Lee, E.; Park, S. E.; Jun, K. Y.; Kadayat, T. M.; Thapa, P.; Karki, R.; Na, Y.; Park, M. S.; Rho, S. B.; Lee, E. S.; Kwon, Y. A series of novel terpyridine-skeleton molecule derivants inhibit tumor growth and metastasis by targeting topoisomerases. *J. Med. Chem.* **2015**, *58*, 1100−1122. (b) Thapa Magar, T. B.; Kadayat, T. M.; Lee, H. J.; Park, S.; Bist, G.; Shrestha, A.; Kwon, Y.; Lee, E. S. 2-Chlorophenyl-substituted benzofuro[3,2-*b*]pyridines with enhanced topoisomerase inhibitory activity: The role of the chlorine substituent. *Bioorg. Med. Chem. Lett.* **2017**, *27*, 3279−3283. (c) Shrestha, A.; Jo, H.; Kwon, Y.; Lee, E. S. Design, synthesis, and structure-activity relationships of new benzofuro[3,2-*b*]pyridin-7-ols as DNA topoisomerase II inhibitors. *Bioorg. Med. Chem. Lett.* **2018**, *28*, 566−571. (d) Thapa Magar, T. B.; Hee Seo, S.; Shrestha, A.; Kim, J. A.; Kunwar, S.; Bist, G.; Kwon, Y.; Lee, E. S. Synthesis and structure-activity relationships of hydroxylated and halogenated 2,4-diaryl benzofuro[3,2-*b*]pyridin-7-ols as selective topoisomerase IIalpha inhibitors. *Bioorg. Chem.* **2021**, *111*, 104884−104898.

(4) Peng, W.; Sun, Z. Y.; Zhang, Q.; Cheng, S. Q.; Wang, S. K.; Wang, X. N.; Kuang, G. T.; Su, X. X.; Tan, J. H.; Huang, Z. S.; Ou, T. M. Design, Synthesis, and Evaluation of Novel p-(Methylthio)styryl Substituted Quindoline Derivatives as Neuroblastoma RAS (NRAS) Repressors via Specific Stabilizing the RNA G-Quadruplex. *J. Med. Chem.* **2018**, *61*, 6629−6646.

(5) Zhou, J.-L.; Lu, Y.-J.; Ou, T.-M.; Zhou, J.-M.; Huang, Z.-S.; Zhu, X.-F.; Du, C.-J.; Bu, X.-Z.; Ma, L.; Gu, L.-Q.; Li, Y.-M.; Chan, A. S.-C. Tricyclic pyrazoles. 3. Synthesis, biological evaluation, and molecular modeling of analogues of the cannabinoid antagonist 8-chloro-1-(2′,4′-dichlorophenyl)-N-piperidin-1-yl-1,4,5,6-tetrah-ydrobenzo-[6,7]cyclohepta [1,2-*c*]pyrazole-3-carboxamide. *J. Med. Chem.* **2005**, *48*, 7315−7362.

(6) For selected examples, see: (a) Yue, W.-S.; Li, J.-J. Diastereoselective Synthesis of 1,2-Dihydrobenzofuro[3,2-*b*]pyridines via a Carbon−Carbon Double-Bond Cleavage/Rearran-gement Cascade. *Org. Lett.* **2002**, *4*, 2201−2203. (b) Liu, J.; Fitzgerald, A.-E.; Mani, N.-S. Facile Assembly of Fused Benzo[4,5]furo Heterocycles. *J. Org. Chem.* **2008**, *73*, 2951−2954. (c) Funicello, M.; Laboragine, V.; Pandolfo, R.; Spagnolo, P. A New Entry to Benzo[4,5]furo[3,2-*b*]pyridines via N-(Benzofuran-3-yl)-iminophosphorane. *Synlett* **2010**, *2010*, 77−80. (d) Sun, W.; Wang, M.; Zhang, Y.; Wang, L. Synthesis of benzofuro[3,2-*b*]pyridines via palladium-catalyzed dual C-H activation of 3-phenoxypyridine 1-oxides. *Org. Lett.* **2015**, *17*, 426−429.

(7) Zeng, R.; Shan, C.; Liu, M.; Jiang, K.; Ye, Y.; Liu, T. Y.; Chen, Y. C. [4 + 1+1] Annulations of α-Bromo Carbonyls and 1-Azadienes toward Fused Benzoazaheterocycles. *Org. Lett.* **2019**, *21*, 2312−2316.

(8) Xie, H. P.; Sun, L.; Wu, B.; Zhou, Y. G. Copper-Catalyzed Alkynylation/Cyclization/Isomerization Cascade for Synthesis of 1,2-Dihydrobenzofuro[3,2-*b*]pyridines and Benzofuro[3,2-*b*]pyridines. *J. Org. Chem.* **2019**, *84*, 15498−15507.

(9) Zhu, C. F.; Chen, L. Q.; Hao, W. J.; Cui, C. C.; Tu, S. J.; Jiang, B. Diastereoselective Synthesis of 1,2-Dihydroben-zofuro[3,2-*b*]-pyridines via a Carbon-Carbon Double-Bond Cleavage/Rearrange-ment Cascade. *Org. Lett.* **2021**, *23*, 2654−2658.

(10) For selected examples, see: (a) Rong, Z. Q.; Wang, M.; Chow, C. H.; Zhao, Y. A Catalyst-Enabled Diastereodivergent Aza-Diels-Alder Reaction: Complementarity of N-Heterocyclic Carbenes and Chiral Amines. *Chem. - Eur. J.* **2016**, *22*, 9483−9487. (b) Chen, J.; Huang, Y. Phosphine-Catalyzed Sequential [4 + 3] Domino Annulation/Allylic Alkylation Reaction of MBH Carbonates: Efficient Construction of Seven-Membered Heterocycles. *Org. Lett.* **2017**, *19*, 5609−5612. (c) Yang, L. C.; Rong, Z. Q.; Wang, Y. N.; Tan, Z. Y.; Wang, M.; Zhao, Y. Construction of Nine-Membered Heterocycles through Palladium-Catalyzed Formal [5 + 4] Cycloaddition. *Angew. Chem., Int. Ed.* **2017**, *56*, 2927−2931. (d) Ni, H.; Tang, X.; Zheng, W.; Yao, W.; Ullah, N.; Lu, Y. Enantioselective Phosphine-Catalyzed Formal [4 + 4] Annulation of alpha,beta-Unsaturated Imines and Allene Ketones: Construction of Eight-Membered Rings. *Angew. Chem., Int. Ed.* **2017**, *56*, 14222−14226. (e) Rong, Z. Q.; Yang, L. C.; Liu, S.; Yu, Z.; Wang, Y. N.; Tan, Z. Y.; Huang, R. Z.; Lan, Y.; Zhao, Y. Nine-Membered Benzofuran-Fused Heterocycles: Enantioselective Synthesis by Pd-Catalysis and Rearrangement via Transannular Bond Formation. *J. Am. Chem. Soc.* **2017**, *139*, 15304−15307. (f) Gu, Z.; Wu, B.; Jiang, G.-F.; Zhou, Y.-G. Synthesis of Benzofuran-fused 1,4-Dihydropyridines via Bifunctional Squaramide-catalyzed Formal [4 + 2] Cycloaddition of Azadienes with Malononitrile. *Chin. J. Chem.* **2018**, *36*, 1130−1134. (g) Gu, Z.; Zhou, J.; Jiang, G.-F.; Zhou, Y.-G. Synthesis of chiral γ-aminophosphonates through the organocatalytic hydrophosphonylation of azadienes with phosphites. *Org. Chem. Front.* **2018**, *5*, 1148−1151. (h) Gu, Z.; Xie, J.-J.; Jiang, G.-F.; Zhou, Y.-G. Catalytic Asymmetric Conjugate Addition of Tritylthiol to Azadienes with a Bifunctional Organocatalyst. *Asian J. Org. Chem.* **2018**, *7*, 1561−1564. (i) Chen, J.; Jia, P.; Huang, Y. Divergent Domino Reactions of Sulfur Ylides: Access to Functionalized Six- and Seven-Membered Nitrogen-Heterocycles. *Org. Lett.* **2018**, *20*, 6715−6718. (j) Gao, Z. H.; Chen, K. Q.; Zhang, Y.; Kong, L. M.; Li, Y.; Ye, S. Enantioselective N-Heterocyclic Carbene-Catalyzed Synthesis of Spirocyclic Oxindole-benzofuroazepinones. *J. Org. Chem.* **2018**, *83*, 15225−15235. (k) Lin, W.; Zhang, C.; Xu, W.; Cheng, Y.; Li, P.; Li, W. Organocatalytic Asymmetric Michael Addition of Rhodanines to Azadienes for Assembling of Sulfur-containing Tetrasubstituted Carbon Stereocenters. *Adv. Synth. Catal.* **2019**, *361*, 476−480. (l) Xie, H.-P.; Wu, B.; Wang, X.-W.; Zhou, Y.-G. Chiral Brønsted acid-catalyzed conjugate addition of indoles to azadienes: Enantioselective synthesis of hetero-triarylmethanes. *Chin. J. Catal.* **2019**, *40*, 1566−1575. (m) Wang, C. S.; Li, T. Z.; Cheng, Y. C.; Zhou, J.; Mei, G. J.; Shi, F. Catalytic Asymmetric [4 + 1] Cyclization of Benzofuran-Derived Azadienes with 3-Chlorooxindoles. *J. Org. Chem.* **2019**, *84*, 3214−3222. (n) Li, X.; Yan, J.; Qin, J.; Lin, S.; Chen, W.; Zhan, R.; Huang, H. Enantioselective Synthesis of Benzofuran-Fused N-Heterocycles via Chiral Squaramide Catalyzed [4 + 2] Cyclization of Azadienes with Azlactones. *J. Org. Chem.* **2019**, *84*, 8035−8045. (o) Fan, T.; Zhang, Z. J.; Zhang, Y. C.; Song, J. Construction of All-Carbon Quaternary Stereocenters via Sequential Photoactivation/Isothiourea Catalysis. *Org. Lett.* **2019**, *21*, 7897−7901. (p) Marques, A. S.; Duhail, T.; Marrot, J.; Chataigner, I.; Coeffard, V.; Vincent, G.; Moreau, X. A Fused Hexacyclic Ring System: Diastereoselective Polycyclization of 2,4-Dienals through an Interrupted iso-Nazarov Reaction. *Angew. Chem., Int. Ed.* **2019**, *58*, 9969−9973. (q) Chen, K.-Q.; Gao, Z.-H.; Ye, S. Bifunctional N-heterocyclic carbene catalyzed [3 + 4] annulation of enals with azadienes: enantioselective synthesis of benzofuroazepinones. *Org. Chem. Front.* **2019**, *6*, 405−409. (r) Lin, W.; Lin, X.; Cheng, Y.; Chang, X.; Zhou, S.; Li, P.; Li, W. Organocatalytic enantioselective direct vinylogous Michael addition of γ-substituted deconjugate butenolides to azadienes. *Org. Chem. Front.* **2019**, *6*, 2452−2456.

(11) For selected examples, see: (a) Hashimoto, T.; Sakata, K.; Tamakuni, F.; Dutton, M. J.; Maruoka, K. Phase-transfer-catalysed asymmetric synthesis of tetrasubstituted allenes. *Nat. Chem.* **2013**, *5*, 240−244. (b) Hu, Y.; Shi, W.; Zheng, B.; Liao, J.; Wang, W.; Wu, Y.; Guo, H. Organocatalytic Asymmetric C(sp^2)-H Allylic Alkylation: Enantioselective Synthesis of Tetrasubstituted Allenoates. *Angew. Chem., Int. Ed.* **2020**, *59*, 19820−19824.

https://doi.org/10.1021/acs.orglett.1c02357
Org. Lett. 2021, 23, 6780−6783